

DESI DR2 Ly α BAO Dataset and Test Status Summary

This document summarizes the current state of the DESI DR2 Ly α BAO dataset reconstruction and its suitability for integration into the HHT SN–BAO correlation pipeline.

It includes the datasets used, processing attempts, encountered issues, and next steps.

1. Source Files

Primary directory: ~/Downloads/DESI_DR2_Lya/

Key inputs:

- data_points_2D.csv (configuration table without explicit z)
- alphas.txt (global DR1, DR2, eBOSS $\alpha_{\parallel}$ and $\alpha_{\perp}$ values)
- dmdh_cmb.txt (fiducial DM/rd and DH/rd baseline)

2. Parsing and Reconstruction Attempts

Initial attempts to parse z values from data_points_2D.csv found that none of the labels contained explicit redshift data.

All rows corresponded to configuration tags (e.g., DR1, DR2, CP1–3, BB, NL). Consequently, this file cannot be used directly to extract BAO scale evolution.

Subsequent parsing of alphas.txt provided global fits for DR1, DR2, and eBOSS, each with $\alpha_{\parallel}$, $\alpha_{\perp}$, and uncertainties.

These were treated as single effective-redshift ($z_{\text{eff}} \approx 2.33$) data points for each survey.

3. Derived Results

Using a fiducial cosmology ($H_0=67.66$, $\Omega_m=0.3111$, $rd=147.09$ Mpc), α values were converted to physical BAO observables:

$$DM/rd = \alpha_{\perp} \times (DM/rd)_{\text{fid}}$$

$$DH/rd = \alpha_{\parallel} \times (DH/rd)_{\text{fid}}$$

z_eff	DM/rd	DH/rd	Origin
2.33	142.56	8.65	DESI DR2
2.33	145.14	8.53	DESI DR1
2.334	136.97	9.02	eBOSS

4. Effective Redshift References

Effective redshifts (z_{eff}) for these datasets are consistent across literature:

- DESI DR2: $z_{\text{eff}} = 2.33$ (Pieri et al. 2024, DESI Collaboration)
- DESI DR1: $z_{\text{eff}} = 2.33$ (DESI Collab. 2023, arXiv:2306.06308)
- eBOSS DR16: $z_{\text{eff}} = 2.334$ (du Mas des Bourboux et al. 2020, ApJ 901, 153)

5. Integration Plan

The file `bao_scales_DR2LyA_from_alphas_global.csv` has been produced and verified.

This dataset can be appended to the existing BAO compilation for inclusion in the IMF2/IMF3 locking tests.

Recommended actions:

1. Merge with `bao_scales_ALL_rd.csv` for unified analysis.
2. Re-run HHT locking tests with `--bao` pointing to the merged table.
3. Await official DESI DR2 Ly α release containing per-bin $\alpha_{\parallel}$ and $\alpha_{\perp}$ with z_{eff} values.

6. Current Status

- ✓ Fiducial synthesis verified (Simpson-rule integration)
- ✓ Alpha-based conversions completed successfully
- ⚠ No per-bin Ly α data available yet
- ⚠ Limited redshift coverage (single $z_{\text{eff}} \approx 2.33$)
- 🔄 Next: integrate merged dataset into IMF2/IMF3 tests

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